

TLC Trip Record Data Processing

Big Data | 2nd Assignment

SPRING 2025

STACK:

Juan Sebastian Vargas Torres

juan.vargastorres@student.uts.edu.au

UTS

Table of Contents

Introduction *3*

Dataset **3**

High level Architecture *3*

Bronze Layer **3**

a. Green Taxi Data 4
b. Yellow Taxi Data 6
c. Location Data 7

Silver Layer **8**

a. Initial Data Loading 8
b. Data Filering and Cleansing 8
c. Data Unification & Enrichment 9

Gold Layer **10**

Business Intelligence Views **10**

1a. For each year and month: What was the total number of trips? 10
1b. For each year and month: Which day of week had the most trips? 10
1c. For each year and month: Which hour of the day had the most trips? 11
1d. For each year and month: What was the average number of passengers? 11
1e. For each year and month: What was the average amount paid per trip (using total_amount)? 11
1f. For each year and month: What was the average amount paid per passenger (using total_amount)? 12
2. For each taxi colour, what were the key trip statistics? 12
3. For each taxi colour and location pair, what were the granular trip statistics? 12
4. For 2024, compute the share of total revenue contributed by the Top 10 pickup→dropoff borough pairs. 13
5. What was the percentage of trips where drivers received tips? 13
6. For trips where the driver received tips, what was the percentage where the driver received tips of at least $15? 13
7. For different trip duration bins, what were the average speed and distance per dollar? 13
8. Which duration bin will you advise a taxi driver to target to maximise his income? 14

ML Models **14**

a. Model Training and Evaluation 15
b. Model Performance Comparison 16
c. Best Model Selection and Final Performance 16

Findings and recommendations *17*

References *17*

Figure 1. High Level Architecture 3
Figure 2. Dictionary Trips Green Taxis 5
Figure 3 Dictionary Trips Yellow Taxis 6
Figure 4. Dictionary Location Data 7

Introduction

The New York City Taxi & Limousine Commission (TLC) is the agency responsible for licensing and regulating New York City's for-hire vehicle industries, including the iconic yellow medallion taxis, green street hail livery vehicles. As part of its commitment to transparency and public data access, the TLC makes detailed trip-level data for these services publicly available.

Dataset

A comprehensive dataset containing individual trip records for vehicles regulated by the TLC has been made available on the official NYC government website. (https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page)

This dataset includes millions of trip records for **yellow taxis** (dating back to 2001) and **green taxis.** The data is published monthly with a two-month delay and is available for download in PARQUET format.

High level Architecture

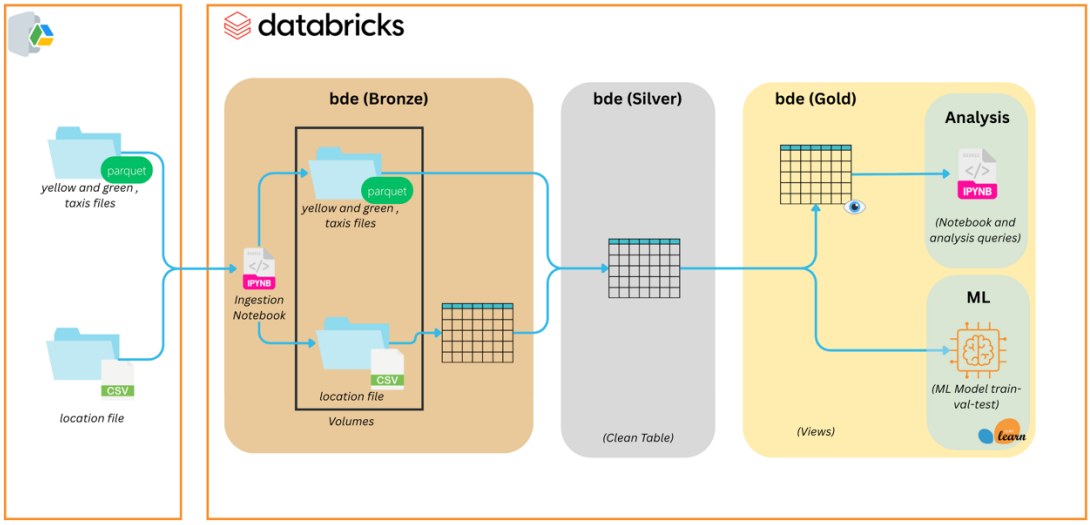


Figure 1. High Level Architecture

Bronze Layer

This layer serves as the initial landing zone for all raw source data. The primary function of the Bronze layer is to ingest and store data in its original, unaltered format. For this project, a dedicated data ingestion notebook was created to fetch the source files for **Yellow Taxis**, **Green Taxis**, and **Location lookups**.

UTS

These files are then stored in the following volume, organized into specific directories for clarity:

workspace.bde.assignment2

- /green
- /yellow
- /location

After successfully storing the files, an Exploratory Data Analysis (EDA) notebook was developed. This EDA process allowed for an initial exploration of the raw data to understand its structure, schema, value distributions, and identify any potential data quality issues before any transformations are applied.

a. Green Taxi Data

This section contains the trip records for green taxis. The data is stored in a Parquet file format.

- **File:** green_taxi.parquet
- **Size:** 1.26 GB
- **Total Records:** 83,484,688

Data Dictionary

Field Name	Description
VendorID	A code indicating the LPEP provider that provided the record. 1 = Creative Mobile Technologies, LLC 2 = Curb Mobility, LLC 6 = Myle Technologies Inc
lpep_pickup_datetime	The date and time when the meter was engaged.
lpep_dropoff_datetime	The date and time when the meter was disengaged.
store_and_fwd_flag	This flag indicates whether the trip record was held in vehicle memory before sending to the vendor.
RatecodeID	The final rate code in effect at the end of the trip. 1 = Standard rate 2 = JFK 3 = Newark 4 = Nassau or Westchester 5 = Negotiated fare 6 = Group ride 99 = Null/unknown
PULocationID	TLC Taxi Zone in which the taximeter was engaged.
DOLocationID	TLC Taxi Zone in which the taximeter was disengaged.
passenger_count	The number of passengers in the vehicle.
trip_distance	The elapsed trip distance in miles reported by the taximeter.
fare_amount	The time-and-distance fare calculated by the meter.
extra	Miscellaneous extras and surcharges.
mta_tax	Tax that is automatically triggered based on the metered rate in use.
tip_amount	Tip amount - This field is automatically populated for credit card tips. Cash tips are not included.
tolls_amount	Total amount of all tolls paid in trip.
improvement_surcharge	Improvement surcharge assessed trips at the flag drop.
total_amount	The total amount charged to passengers. Does not include cash tips.
payment_type	A numeric code signifying how the passenger paid for the trip. 0 = Flex Fare trip 1 = Credit card 2 = Cash 3 = No charge 4 = Dispute 5 = Unknown 6 = Voided trip
trip_type	A code indicating whether the trip was a street-hail or a dispatch. 1 = Street-hail 2 = Dispatch
congestion_surcharge	Total amount collected in trip for NYS congestion surcharge.
cbd_congestion_fee	Per-trip charge for MTA's Congestion Relief Zone starting Jan. 5, 2025.

Figure 2. Dictionary Trips Green Taxis

Key Insights

- **Missing Data:** There are significant amounts of missing data in columns like congestion_surcharge (only ~9M records out of ~83M), improvement_surcharge (~69M), and trip_type (~80M).
- **Extreme Outliers & Errors:** The trip_distance and financial columns (fare_amount, total_amount, etc.) contain impossible values. For example, a minimum trip_distance of -20,329 miles and a maximum of 360,068 miles are clearly data

errors. Similarly, negative fare and tip amounts indicate erroneous records that must be cleaned.

- **Implausible Passenger Counts:** While the average passenger_count is reasonable, the maximum value of 48 is not possible for a standard taxi and represents a significant outlier. Records with 0 passengers may also need investigation.
- **High Variance:** The extremely large standard deviation for trip_distance and total_amount relative to their means confirms the presence of extreme outliers that are skewing the overall data distribution.

b. Yellow Taxi Data

This section contains the trip records for yellow taxis. This is the largest dataset, stored in Parquet format.

- **File:** yellow_taxi.parquet
- **Size:** 11.78 GB
- **Total Records:** 907,982,776

Data Dictionary

Field Name	Description
VendorID	A code indicating the TPEP provider that provided the record. 1 = Creative Mobile Technologies, LLC 2 = Curb Mobility, LLC 6 = Myle Technologies Inc 7 = Helix
tpep_pickup_datetime	The date and time when the meter was engaged.
tpep_dropoff_datetime	The date and time when the meter was disengaged.
passenger_count	The number of passengers in the vehicle.
trip_distance	The elapsed trip distance in miles reported by the taximeter.
RatecodeID	The final rate code in effect at the end of the trip. 1 = Standard rate 2 = JFK 3 = Newark 4 = Nassau or Westchester 5 = Negotiated fare 6 = Group ride 99 = Null/unknown
store_and_fwd_flag	This flag indicates whether the trip record was held in vehicle memory before sending to the vendor. Y = store and forward trip N = not a store and forward trip
PULocationID	TLC Taxi Zone in which the taximeter was engaged.
DOLocationID	TLC Taxi Zone in which the taximeter was disengaged.
payment_type	A numeric code signifying how the passenger paid for the trip. 0 = Flex Fare trip 1 = Credit card 2 = Cash 3 = No charge 4 = Dispute 5 = Unknown 6 = Voided trip
fare_amount	The time-and-distance fare calculated by the meter.
extra	Miscellaneous extras and surcharges.[1]
mta_tax[2]	Tax that is automatically triggered based on the metered rate in use.[3]
tip_amount[4]	Tip amount - This field is automatically populated for credit card tips. Cash tips are not included.[5]
tolls_amount[6]	Total amount of all tolls paid in trip.
improvement_surcharge	Improvement surcharge assessed at the flag drop.
total_amount	The total amount charged to passengers. Does not include cash tips.
congestion_surcharge	Total amount collected in trip for NYS congestion surcharge.
airport_fee	For pick up only at LaGuardia and John F. Kennedy Airports.
cbd_congestion_fee	Per-trip charge for MTA's Congestion Relief Zone starting Jan. 5, 2025.

Figure 3 Dictionary Trips Yellow Taxis

Key Insights

- **Severe Data Corruption:** The trip_distance and financial columns (fare_amount, tip_amount, total_amount) show nonsensical minimum and maximum values (e.g., min fare of -$133 million, max tip of $133 million).
- **Missing Surcharge/Fee Data:** Similar to the green taxi data, the congestion_surcharge and airport_fee columns have far fewer records than the total, indicating they are not universally applied or were introduced partway through the data collection period.
- **Impossible Passenger Values:** The maximum passenger_count of 208 is physically impossible for a taxi, highlighting another clear data outlier.

c. Location Data

This is a supplementary lookup table used to map LocationID values from the trip records to their corresponding borough and zone names. It is stored as a CSV file.

- **File:** taxi_zone_lookup.csv
- **Size:** 12.03 KB

Data Dictionary

Column Name	Data Type	Description
LocationID	Integer	A unique number identifying each location zone.
Borough	String	The New York City borough where the zone is located.
Zone	String	The name of the specific neighborhood or area.
service_zone	String	The category of taxi service for that zone (e.g., Boro Zone, Yellow Zone).

Figure 4. Dictionary Location Data

Key Insights

- **Completeness:** The dataset is perfectly clean with **no missing values**, as all 265 records have data for every column.
- **Unique Identifiers:** There are 265 unique taxi zones, with LocationIDs ranging sequentially from 1 to 265. This file acts as a dimension table to enrich the main trip data.

- **Potential Data Anomaly:** The Borough column contains an 'Unknown' value. This may represent out-of-borough locations (like Newark Airport) or could be a minor data quality issue that needs to be handled.

Silver Layer

This layer takes two separate and messy datasets (for green and yellow taxis) and turns them into a single, trustworthy table. This clean table provides a solid foundation for tasks like spotting trends, understanding fares, or making predictions.

a. Initial Data Loading

The process begins by loading the raw green and yellow taxi datasets from their Bronze Layer storage locations.

b. Data Filering and Cleansing

Cleaning Step	Goal	Action
Validate Trip Chronology	Ensure the trip ends after it starts.	Remove records where the dropoff time is before the pickup time.
Filter for Realistic Dates	Keep only valid historical dates.	Remove records with pickup or dropoff years later than 2024.
Ensure Valid Speed	Remove trips with impossible speeds.	Calculate trip duration and speed, then keep only trips with a speed greater than zero.
Filter by Speed Limits	Eliminate unrealistic speeds based on taxi type.	Filter green taxi trips for speeds 30 mph and yellow taxi trips for speeds 25 mph.
Filter by Trip Duration & Distance	Remove outliers in trip length or distance.	Keep trips with a duration between 1 minute and 3 hours, and a distance between 0.1 and 40 miles.
Drop High-Null Columns	Remove columns with excessive missing values.	Drop columns like ehail_fee (green taxi) or airport_fee (yellow taxi), and congestion_surcharge.

Cleaning Step	Goal	Action
Drop Uninformative Columns	Remove columns that don't add analytical value.	Drop the improvement_surcharge column from both datasets.
Clean Core Trip Data	Ensure valid passenger counts and no nulls in key fields.	Remove trips with 0 passengers, then drop rows with any remaining null values in important fields.
Impute Missing Values	Fill in missing categorical data.	Fill null values in trip_type and store_and_fwd_flag with the most common value (mode).

c. Data Unification & Enrichment

Combine datasets & add location data

- **Goal**: Create one complete table with all taxi trips.
- **Action**: Rename datetime columns for consistency, add a taxi_type column, and merge both datasets. Join the combined data with location info (borough and zone names).

Apply final filters

- **Goal**: Perform last checks on the data.
- **Action**: Remove records with unknown boroughs, negative amounts, or more than 6 passengers.

Finally, the cleaned df was saved as a Delta table in the Silver Layer, ready to use.

Gold Layer

Business Intelligence Views

This part of the Gold Layer focuses on creating pre-aggregated, business-centric SQL views for easy analysis.

1a. For each year and month: What was the total number of trips?

- **View Name:** trips_per_month

	trip_month	total_trips
1	2024-12-01	3084922
2	2024-11-01	3027957
3	2024-10-01	3175012
4	2024-09-01	2911619
5	2024-08-01	2480689
6	2024-07-01	2552495
7	2024-06-01	2897744
8	2024-05-01	3082795
9	2024-04-01	2861173
10	2024-03-01	2909090
11	2024-02-01	2607962
12	2024-01-01	2585761
13	2023-12-01	2957497
14	2023-11-01	2961179
15	2023-10-01	3105045

1b. For each year and month: Which day of week had the most trips?

- **View Name:** top_weekdays

	trip_month	weekday	total_trips
1	2024-12-01	Tuesday	483085
2	2024-11-01	Saturday	539915
3	2024-10-01	Thursday	551843
4	2024-09-01	Thursday	447006
5	2024-08-01	Thursday	449305
6	2024-07-01	Wednesday	465827
7	2024-06-01	Saturday	479494
8	2024-05-01	Thursday	558606
9	2024-04-01	Tuesday	481665
10	2024-03-01	Friday	499487
11	2024-02-01	Thursday	518876
12	2024-01-01	Wednesday	441390
13	2023-12-01	Friday	528760
14	2023-11-01	Wednesday	524240
15	2023-10-01	Tuesday	496335

1c. For each year and month: Which hour of the day had the most trips?

- **View Name:** top_hours

	trip_month	trip_hour	total_trips
1	2024-12-01	18	224808
2	2024-11-01	18	225126
3	2024-10-01	18	238530
4	2024-09-01	18	216068
5	2024-08-01	18	182604
6	2024-07-01	18	187231
7	2024-06-01	18	211468
8	2024-05-01	18	227176
9	2024-04-01	18	209395
10	2024-03-01	18	208686
11	2024-02-01	18	199730
12	2024-01-01	18	190913
13	2023-12-01	18	209050
14	2023-11-01	18	220194
15	2023-10-01	18	227658

1d. For each year and month: What was the average number of passengers?

- **View Name:** avg_passagers_per_month

	trip_month	avg_pass
1	2024-12-01	1.37
2	2024-11-01	1.34
3	2024-10-01	1.32
4	2024-09-01	1.31
5	2024-08-01	1.36
6	2024-07-01	1.36
7	2024-06-01	1.35
8	2024-05-01	1.34
9	2024-04-01	1.35
10	2024-03-01	1.35
11	2024-02-01	1.34
12	2024-01-01	1.35
13	2023-12-01	1.42
14	2023-11-01	1.37
15	2023-10-01	1.38

1e. For each year and month: What was the average amount paid per trip (using total_amount)?

- **View Name:** avg_amount_per_month

	trip_month	avg_amount
1	2024-12-01	27.29
2	2024-11-01	26.6
3	2024-10-01	27.43
4	2024-09-01	27.8
5	2024-08-01	26.7
6	2024-07-01	26.44
7	2024-06-01	26.83
8	2024-05-01	27.13
9	2024-04-01	26.08
10	2024-03-01	25.7
11	2024-02-01	24.76
12	2024-01-01	24.21
13	2023-12-01	26.7
14	2023-11-01	26.83
15	2023-10-01	27.24

1f. For each year and month: What was the average amount paid per passenger (using total_amount)?

- **View Name:** avg_amount_per_pass

	trip_month	avg_amount_per_pass
1	2024-12-01	23.33
2	2024-11-01	23.1
3	2024-10-01	23.93
4	2024-09-01	24.36
5	2024-08-01	22.84
6	2024-07-01	22.72
7	2024-06-01	23.22
8	2024-05-01	23.59
9	2024-04-01	22.62
10	2024-03-01	22.3
11	2024-02-01	21.64
12	2024-01-01	21.1
13	2023-12-01	22.48
14	2023-11-01	23.1
15	2023-10-01	23.42

2. For each taxi colour, what were the key trip statistics?

- **View Name:** taxis_colour_distance_duration_speed_stats

taxi_type	avg_duration_min	median_duration_min	min_duration_min	max_duration_min	avg_distance_km	median_distance_km	min_distance_km
green	13.44	10.42	1	180	4.55	3.06	0.16
yellow	13.93	10.95	1	180	4.19	2.66	0.16

max_distance_km	avg_speed_kmh	median_speed_kmh	min_speed_kmh	max_speed_kmh
64.37	20.32	18.39	0.06	48.28
64.37	18.02	16.09	0.05	40.23

3. For each taxi colour and location pair, what were the granular trip statistics?

- **View Name:** taxis_colour_pickup_dropoff_stats

taxi_type	pickup_borough	dropoff_borough	month(pickup_datetime)	weekday	hour(pickup_datetime)	total_trips	avg_distance	avg_amount	sum_total_amount
yellow	Manhattan	EWR	11	Monday	7	216	17.36	95.98	20730.73
yellow	Brooklyn	Manhattan	11	Tuesday	6	2817	5.71	24.73	69659.97
yellow	Bronx	Manhattan	11	Wednesday	6	255	9.23	36.63	9341.71
yellow	Bronx	Bronx	11	Wednesday	7	281	2.71	16.54	4646.67
yellow	Bronx	Bronx	11	Wednesday	17	221	2.8	18.04	3987.59
yellow	Brooklyn	Manhattan	11	Friday	1	1098	4.9	21.92	24068.35
yellow	Brooklyn	Manhattan	12	Thursday	1	881	4.95	21.81	19214.39
yellow	Manhattan	Bronx	12	Saturday	16	1271	7.63	30.61	38905.2
yellow	Queens	Staten Island	10	Friday	17	46	27.07	102.26	4704.15
yellow	Manhattan	Manhattan	11	Wednesday	20	537367	1.95	14.03	7540888.61
yellow	Manhattan	Bronx	11	Thursday	14	1278	8.51	35.69	45606.4
yellow	Brooklyn	EWR	11	Friday	14	6	17.93	116.92	701.53
yellow	Manhattan	Queens	12	Tuesday	15	18764	12.22	54.36	1020017.04
yellow	Queens	Manhattan	12	Sunday	1	1048	5.58	25.2	26414.39
yellow	EWR	EWR	11	Sunday	20	3	0.69	46.77	140.3
yellow	Brooklyn	Brooklyn	12	Friday	3	2502	2.24	11.99	30008.72
yellow	Queens	EWR	12	Monday	18	6	26.51	132.57	795.43

4. For 2024, compute the share of total revenue contributed by the Top 10 pickup→dropoff borough pairs.

- **View Name:** top_pickup_dropoff_pairs_revenue_2024

	pickup_borough	dropoff_borough	pair_total_amount	revenue_share_percent
1	Manhattan	Manhattan	632193530.83	69.9
2	Queens	Manhattan	119937624.17	13.26
3	Manhattan	Queens	51540580.56	5.7
4	Manhattan	Brooklyn	30572977.96	3.38
5	Queens	Brooklyn	24311709.38	2.69
6	Queens	Queens	20888891.75	2.31
7	Brooklyn	Brooklyn	4876906.41	0.54
8	Manhattan	EWR	4621222.95	0.51
9	Brooklyn	Manhattan	4201664.21	0.46
10	Manhattan	Bronx	3996270.06	0.44

5. What was the percentage of trips where drivers received tips?

- **View Name:** percentage_of_trips_with_tips

	percentage_with_tips
1	63.07

6. For trips where the driver received tips, what was the percentage where the driver received tips of at least $15?

- **View Name:** percentage_of_trips_with_at_least_15

	percentage_with_tips
1	0.48

7. For different trip duration bins, what were the average speed and distance per dollar?

- **View Name:** duration_bins_stats

	duration_bin	average_speed_kph	average_km_per_dollar
1	a. Under 5 Mins	18.87	0.16
2	b. 5 to 10 mins	16.83	0.21
3	c. 10 to 20 mins	16.64	0.25
4	d. 20 to 30 mins	18.02	0.29
5	e. 30 to 60 mins	21.8	0.34
6	f. At least 60 mins	21.41	0.49

8. Which duration bin will you advise a taxi driver to target to maximise his income?

- **View Name:** duration_bins_profitability

	taxi_type	duration_bin	income_per_hour	average_fare
1	green	a. Under 5 Mins	108.41	6.46
2	green	b. 5 to 10 mins	74.85	9.23
3	green	c. 10 to 20 mins	64.9	15.29
4	green	d. 20 to 30 mins	60.85	24.48
5	green	e. 30 to 60 mins	56.8	36.77
6	green	f. At least 60 mins	47.25	60.36
7	yellow	a. Under 5 Mins	124.06	7.35
8	yellow	b. 5 to 10 mins	83.06	10.28
9	yellow	c. 10 to 20 mins	66.41	15.64
10	yellow	e. 30 to 60 mins	65.88	43.54
11	yellow	d. 20 to 30 mins	62.18	24.93
12	yellow	f. At least 60 mins	56.29	68.61

- **Suggestion for Green Taxi Drivers**

To maximize their income, Green Taxi drivers should focus on finding as many short-duration fares as possible.
Their most profitable trips are those under 5 minutes, earning them $108.48 per hour. Their profitability drops with every subsequent duration bin, with long trips of 60+ minutes being the least profitable at only $47.73 per hour.

- **Suggestion for Yellow Taxi Drivers**

Yellow Taxi drivers should follow the same strategy of targeting quick, short-distance trips. The "Under 5 Mins" bin is their most lucrative category by a large margin, generating $124.13 per hour. Similar to Green Taxis, their hourly income rate declines as trips get longer, with the lowest rate coming from trips that last over an hour.

ML Models

This section outlines the process of developing a machine learning model to predict the total_amount of a taxi trip.

Data Preparation and Sampling:

- The clean dataset (taxis_trips) was loaded from the Silver Layer.
- An initial filter was applied to select data exclusively from January 1, 2024, onwards.
    - Train: 2024-01-01 → 2024-07-01
    - Validation: 2024-08-01 → 2024-09-01
    - Test: > 2024-09-01

Feature Selection:

- A correlation matrix and Welch's ANOVA test were used to identify and remove redundant or uninformative features.

- Highly correlated features (e.g., duration_in_hours, speed) and categorical variables with low predictive power (e.g., pickup_zone) were dropped.

Feature Engineering:

- The trip data was joined with the taxis_colour_pickup_dropoff_stats Gold view to add the historical average amount (avg_amount) for trips sharing the same time, taxi type, pickup, and dropoff locations. This value was used as the prediction for the Baseline Model.

Data Preprocessing and Splitting:

- **Encoding:** Categorical features like taxi_type were one-hot encoded to be used in the models.
- **Scaling:**
  - The features for the **Lasso Regression** model were scaled using StandardScaler.
  - The data for the **Decision Tree Regressor** was left unscaled, as tree-based models are not sensitive to feature scaling.

- **Data Splitting:** The data was split into time-based training, validation, and test sets to simulate a real-world prediction scenario (**Due to computational constraints, a 30% sample of the data was used from the training, validation, and test sets).**

a. Model Training and Evaluation

Models Evaluated: Three distinct models were trained and compared:

1. **Baseline Model:** Predicts the historical average fare for a given route.
2. **Lasso Regression:** A linear model with L1 regularization.
3. **Decision Tree Regressor:** A non-linear, tree-based model.

Training & Validation:

- **Period:** Models were trained and validated using data from January 1, 2024, to September 30, 2024.
- **Hyperparameter Tuning:** GridSearchCV was used to find the optimal hyperparameters for the Lasso and Decision Tree models.
- **Assessment Metric: Root Mean Squared Error (RMSE)** was the primary metric used to evaluate and compare model performance.

b. Model Performance Comparison

The table below summarizes the RMSE for each model on the training and validation sets.

Model Type	Hyperparameters	Training RMSE	Validation RMSE	Difference (Train-Val)
Baseline Model	N/A	22.8606	30.5321	7.6715
Lasso Regression	alpha=0.1	4.5364	5.1085	0.5720
	alpha=0.25	4.5647	5.1455	0.5808
	alpha=0.5	4.6576	5.2533	0.5957
Decision Tree Regressor	max_depth=25, min_samples_leaf=8	2.9187	3.6580	0.7392
	max_depth=15, min_samples_leaf=4	3.1093	3.7636	0.6542
	max_depth=8, min_samples_leaf=4	**3.7206**	**4.2036**	**0.4830**

c. Best Model Selection and Final Performance

Model Selection: The selected model is the **Decision Tree Regressor with max_depth=8 and min_samples_leaf=4**.

Justification: Although the max_depth=25 model achieved the lowest validation RMSE, it showed a larger gap between its training and validation performance, suggesting a higher risk of overfitting. The chosen model (max_depth=8) provides the best balance, demonstrating the most consistent performance with the smallest difference (0.4830) between training and validation error.

Performance on Test Set: After being trained on the combined training and validation data, the selected model's final performance was evaluated on the held-out test set.

- **Best Model RMSE on Test Set: 4.2557**

This result confirms that the model significantly outperforms the baseline, providing a much more accurate prediction of the total trip amount than simply relying on historical averages.

Findings and recommendations

- **Prioritize Trip Volume Over Duration:** The most effective driver strategy is to maximize the **number of short fares** initiated rather than the fare amount of a single long trip. The data shows that short trips (under 5 minutes) are significantly more profitable per hour than long trips.

- **Optimize the Queens-to-Manhattan Corridor:** As the second most profitable route, accounting for **13.26%** of top-pair revenue, it represents a key market. The TLC should focus on improving service on this route, such as ensuring higher driver availability in key Queens neighborhoods during peak times.

- **Focus Resources on the 6 PM Peak:** The analysis consistently identifies **6 PM** as the daily peak hour for trip volume. The TLC could implement dynamic incentives to encourage more drivers to be active during this window, ensuring supply meets the consistent surge in demand.

- **Enhance Model Accuracy with More Data:** To create a more robust predictive model for a production environment, it is highly recommended to fine-tune it with a larger dataset (e.g., **50% or 100%** of the available data). This would allow the model to learn from more diverse trip scenarios and improve its overall accuracy. However, the current state of the model is good enough to be deployed to a production environment, considering that the sample data used is sufficiently representative of the whole dataset.

References

- TLC Trip Record Data - TLC. (n.d.). NYC.gov. Retrieved September 22, 2025, from https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page